

|             |                            |             |                         |
|-------------|----------------------------|-------------|-------------------------|
| Report ID:  | RPT-E232E940               | Generated:  | 2026-04-12 14:33:01 IST |
| Patient:    | jo                         | Study Date: | April 12, 2026          |
| Patient ID: | 00                         | Modality:   | MRI Brain (AI Inferred) |
| Model:      | BrainTumorAI v2.0 Ensemble | Model AUC:  | 0.9996   Acc 98.78%     |

PRIMARY FINDING

NO TUMOR

Diagnosis: NO TUMOR

Confidence:  40.9%

Risk Level:  LOW

Processing: 4.16s (TTA 6-view ensemble)

Method: EfficientNet-B3 + ResNet50+CBAM + DenseNet121

DIFFERENTIAL DIAGNOSIS

| Class      | Probability | Bar Chart   | Notes     |
|------------|-------------|-------------|-----------|
| Glioma     | 15.3%       | <div></div> |           |
| Meningioma | 29.8%       | <div></div> |           |
| No Tumor   | 40.9%       | <div></div> | ← Primary |
| Pituitary  | 14.1%       | <div></div> |           |

ENSEMBLE AGREEMENT

| Model          | Confidence | Bar         |
|----------------|------------|-------------|
| ResNet50+CBAM  | 40.9%      | <div></div> |
| Ensemble (TTA) | 40.9%      | ← FINAL     |

## IMAGING EVIDENCE

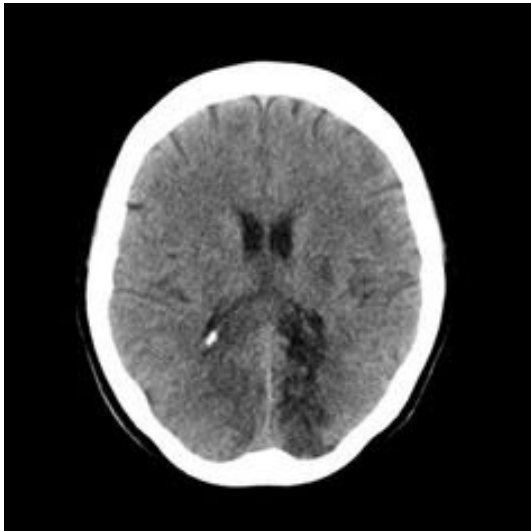


Figure 1: Original MRI scan

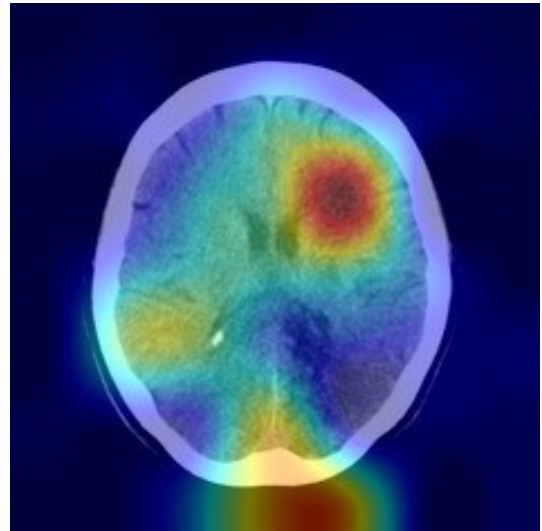


Figure 2: Grad-CAM activation overlay

### GRAD-CAM ANALYSIS PANEL

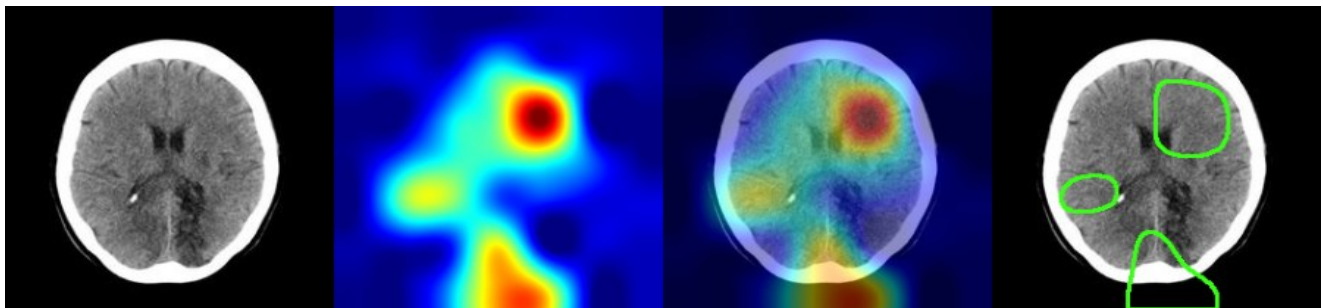


Figure 3: (Left→Right) Original MRI | Activation Map (Jet) | GradCAM Overlay | Tumour Boundary (threshold > 0.5)

■ Red areas = high tumour probability ■ Yellow = moderate ■ Blue = low activation

### TUMOUR LOCALISATION (GRAD-CAM ANALYSIS)

| Parameter             | Value                 | Notes                                       |
|-----------------------|-----------------------|---------------------------------------------|
| Estimated Area        | 11.4% of visible scan | Moderate                                    |
| Hemisphere / Location | CENTRAL               | Based on centroid x-position                |
| Centroid Position     | (123, 148) pixels     | Hottest activation pixel centroid           |
| Bounding Region       | [49,58] → [169,235]   | Pixel coordinates of activation box         |
| Activation Score      | 66 / 100              | Mean activation in region (0=low, 100=high) |
| GradCAM Computed      | 1.19s                 | CPU inference time                          |

## CLINICAL INTERPRETATION

No significant intracranial mass lesion identified by AI analysis. The Grad-CAM map shows no focal area of high activation consistent with tumour pathology. Routine clinical follow-up is advised as clinically indicated. If symptoms persist or new neurological signs develop, repeat imaging should be considered. This result does not exclude all pathologies — clinical correlation is essential.

## RECOMMENDATIONS

- Correlate with clinical history and neurological examination
- No urgent imaging follow-up required based on AI screening
- Repeat MRI in 6–12 months if symptoms persist
- Consider EEG if seizures are present

## TECHNICAL DETAILS

| Parameter       | Value                                             |
|-----------------|---------------------------------------------------|
| Architecture    | EfficientNet-B3 + ResNet50+CBAM + DenseNet121     |
| Inference       | TTA (6-view: orig, H-flip, V-flip, rot90/180/270) |
| Ensemble method | Confidence-weighted average + temperature scaling |
| GradCAM method  | Standard Grad-CAM on last conv layer per backbone |
| Model AUC       | 0.9996 (macro one-vs-rest, test set n=1311)       |
| Model Accuracy  | 98.78%                                            |
| Sensitivity     | 98.68%   Specificity: 99.59%                      |
| Device          | CPU (no GPU detected)                             |
| Processing time | 4.16s (inference) + 1.19s (GradCAM)               |

■ **DISCLAIMER:** This report is generated by NeuroScan AI, an AI-assisted screening tool. It is intended for informational and research purposes only and does **NOT** constitute a medical diagnosis. All findings must be reviewed and confirmed by a qualified radiologist or neurologist before any clinical decision is made. Individual model performance may vary across patient populations and imaging protocols. Reported metrics: AUC 0.9996, Accuracy 98.78%, Sensitivity 98.68%, Specificity 99.59% (test set n=1,311 images). Report generated: 2026-04-12 14:33:01 IST | Report ID: RPT-E232E940